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Is AI my co-author? The ethics of using artificial intelligence in scientific publishing

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ABSTRACT

The recent emergence of Large Language Models (LLMs) and other forms of Artificial Intelligence (AI) has led people to wonder whether they could act as an author on a scientific paper. This paper argues that AI systems should not be included on the author by-line. We agree with current commentators that LLMs are incapable of taking responsibility for their work and thus do not meet current authorship guidelines. We identify other problems with responsibility and authorship. In addition, the problems go deeper as AI tools also do not write in a meaningful sense nor do they have persistent identities. From a broader publication ethics perspective, adopting AI authorship would have detrimental effects on an already overly competitive and stressed publishing ecosystem. Deterrence is possible as backward-looking tools will likely be able to identify past AI usage. Finally, we question the value of using AI to produce more research simply for publication's sake.

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1. Introduction

New questions about the ethics of using artificial intelligence emerge as people consider how to best use these resources in their research practices. Is it possible to ethically use AI tools based on large language models (LLMs) (such as ChatGPT) to draft scientific research papers and if so, do they deserve credit as a coauthor (Kaebrick et al. 2023; Hosseini, Rasmussen, and Resnik 2023; Ritz 2023)? This paper explores the ethical issues of using new artificial intelligence products in manuscript writing. We first consider some of the existing arguments disqualifying LLMs as scientific authors because they lack the ability to take responsibility for their work as many authorship accounts require. Responsibility is not the only problem with AI tools meeting current authorship guidelines, however. In addition, they are unable to write in a scientifically relevant sense and lack meaningful persistent identities over time.

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Authorship guidelines are designed to meet the needs of the current research environment. Should the potential of AI to transform scientific work lead us to change our accounts of authorship to allow for their work? We argue against this idea. First, current versions of LLM-based tools are unable to give fair credit to the ideas used as input and thereby are incompatible with its use in a scientific context. Second, the adoption of these tools for writing would put tremendous strain on the time of peer reviewers, who have no moral obligation to act as peer reviewers for AI-generated papers. Third, embracing AI as a scientific author would unjustly harm researchers who choose not to use AI, by putting them at a competitive disadvantage. We next consider whether it is possible to deter AI use in academic papers. Finally, we end by discussing the value of research and the desirability of adopting AI-enhanced work.

2. Generative AI

Most of the AI tools under consideration in this paper are based on “large language models” (LLMs) in which computer systems have been trained on huge data sets of written language and their task is to predict the next likely word in a sentence given a series of prompts (Shanahan 2024). These LLMs are embedded in an ever-growing number of different AI tools and chatbots (such as ChatGPT); when referring to AI (including chatbots and other AI tools) throughout this paper, we are referring to those systems based on LLMs. While AI tools can be used for a variety purposes, such as producing pictures and videos or providing interactive customer service, in this paper we focus on the use of LLM-based tools to generate written text.

It is important, for the purposes of this paper, to consider two key aspects of any LLM. First, the model is only as good as its algorithms and training set. The training sets are of particular importance, as they provide insight into whose intellectual property was used to train a given model. The other major consideration is that these LLMs lack any and all content knowledge about the domains under consideration (Shanahan 2024). In many ways, these AI models are better thought of as confabulation engines mimicking common human speech (Bender et al. 2021). Just as patients with brain injury will tell a coherent but false story about what they perceived in the absence of any data, these AI tools are merely creating plausible sounding speech. That is not to say that LLMs are not becoming startlingly good at mimicking human writing. Each subsequent update to the LLMs brings them ever closer to modeling competent human writing.

3. Current guidance and ethical issues

In response to the November 2022 launch of ChatGPT, which enabled the generation of longer, more natural-seeming passages of text with the simple

input of a prompt, journals and publishers began updating their publication guidelines to address the use of AI in manuscript submissions. A recent study by Lund and Naheem (2024) found that over half of the top 300 academic journals had some form of policy or guidance regarding the use of AI in journal submissions as of the beginning of April 2023. Of those journals with AI guidelines, nearly 97% allowed some use of AI in manuscript preparation (such as to improve the readability of the paper), but almost all these journals (close to 99%) explicitly forbade granting AI authorship (Lund and Naheem 2024, 14).

Answering the question of whether AI can be a scientific author first requires some clarity on what makes a contribution to a scientific paper worthy of authorship. There is some disagreement in the academic literature about what kinds of contributions are worthy of credit as an author on a journal article or other publication. Many journals follow the International Committee of Medical Journal Editors' (ICMJE) guidance on authorship, which states that authors need to meet four necessary criteria to count as an author. The ICMJE requires the following of would-be authors:

- “Substantial contributions to the conception or design of the work; or the acquisition, analysis, or interpretation of data for the work; AND
- Drafting the work or reviewing it critically for important intellectual content; AND
- Final approval of the version to be published; AND
- Agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved” (ICMJE 2023).

As many commentators have noted, no current form of AI can meet these conditions (Kaechnick et al. 2023; Hosseini, Rasmussen, and Resnik 2023). For instance, there is no robust sense in which current iterations of AI can grant final approval of the manuscript. Similarly, authors need to consent to publication, and are usually asked to disclose any conflicts of interest. AI tools simply cannot legally (or morally) consent or attest to anything (COPE 2023; Zielinski et al. 2023).

Nor is it likely for any AI to be able to take “responsibility” anytime soon, and thus it cannot satisfy the fourth authorship criterion. Chatbots and other AI tools cannot explain or justify their work and cannot realistically be punished for any instances of research misconduct, such as fabrication or plagiarism, that they engaged in (Hosseini, Rasmussen, and Resnik 2023). They cannot be put on trial and cannot lose grant funding, for instance. At a stretch, we might say that a chatbot could lose its “job” through banning its use by institutions or human coauthors that oversaw the research or

submitted the work, but this would more precisely be punishing the companies that produced these tools rather than the tools themselves.¹

Intuitively, most people would hesitate to ascribe ethical concepts like responsibility and agency to AI systems. Of course, these types of intuitions may change if we are at the dawn of a revolutionary AI era. As things currently stand, though, AI can never count as an author according to the ICMJE criterion and any similar models of authorship that also require the acceptance of moral accountability (Tamburrini 2022).

Of course, to ban AI authorship is not to ban all uses of AI in the development of a manuscript, and most current guidelines do permit the use of AI in at least some aspects of research and manuscript preparation as long as this usage is properly disclosed, usually either in the acknowledgments or methods section of the paper (Lund and Naheem 2024). In terms of how AI may be used, some journals, such as Elsevier, permit it only for improving the readability of manuscripts or processing data, not for generating original content (Elsevier 2023; see also Thorp 2023), while others are expansive in allowing (either explicitly or through a lack of stipulated restrictions) AI to be used in any aspect of research or manuscript preparation.

4. Publication ethics issues with AI authorship

Clearly, the current consensus on scientific authorship forbids awarding authorship credit to an AI tool on a research paper. However, even while explicitly barring AI authorship, expansive or unrestricted AI policies nonetheless tacitly allow AI tools to be used in ways that could otherwise qualify for authorship, were a human (with the ability to be responsible and accountable) to make these contributions. Some have argued against using the lack of responsibility or accountability as a reason to deny authorship to AI. For instance, Jenkins and Lin (2023) state that the practice of granting posthumous authorship indicates that we do not consistently expect authors to be accountable for their work. If this is not used as a reason to deny posthumous authorship, can it consistently be used as a reason to deny AI authorship? We will argue that there is an important difference between *taking* responsibility for an existing work (which is required by many current guidelines, and which neither posthumous authors nor AI tools are able to do) and *being* responsible for the creation of a work, which (so far) only humans can do. Because of this, AI tools and deceased writers are not equivalent in regard to every aspect of responsibility.

Whether the authorship guidelines should be changed to enable AI authorship depends in part, then, on whether and how responsibility should rightfully be considered necessary for authorship. If we allow ourselves to move beyond mere compliance with existing guidelines, we can consider if

the guidelines themselves need to be modified. Authorship guidelines set out what is required to count as an author in the current research climate. These guidelines can and do change. For example, the requirement that authors take accountability and/or responsibility for their work is a relatively new addition. The ICMJE criterion grew out of a 1978 meeting in Vancouver to create uniform manuscript guidelines (ICMJE 2024). These conditions were likely added in response to people unethically claiming authorship credit when they did not deserve it.

Some have argued, though, that the ICMJE account of authorship is too restrictive (Moffatt 2013). It tries to make many unethical practices like honorary authorship (where a famous person is added as an author to a paper on which they did nothing to give the paper added prestige) impossible by explicitly creating additional conditions that make this kind of contribution not worthy of authorship. This attempt to redefine authorship as ethical authorship conflates two distinct questions: what makes someone an author and did the authors act ethically (Moffatt 2013).

If we relax the authorship standards, should AI count as an author? Being an author is not simply about the production of words. We argue that responsibility is not just a matter of having someone who can answer for the research after publication (in other words, having someone who takes responsibility for the work); instead, responsibility is also an important part of authorship itself. Thus, although an argument can be made against including a separate responsibility criterion in authorship guidelines, there is some aspect of responsibility that forms an intrinsic part of authorship itself. It is this aspect of responsibility (being responsible for the creation of a work) that marks an important distinction between AI-generated and posthumous authorship.

5. Broader philosophical concerns about AI authorship and responsibility in research

In discussions of the need (or lack thereof) for responsibility and accountability in authorship, it can often appear that these are concerns that solely arise *after* manuscript submission, and so are concomitant to but not part of authorship itself. One authors the work, and then, if any aspect of it is later called into question, one must be able to explain or justify the work (and may be punished in some way if the explanation is lacking). If responsibility merely *followed* authorship in this way, the example of posthumous authorship would rightly call into question the restriction of authorship to those that can be held both morally and legally responsible for their work. However, we argue that, rather than simply following from authorship, some aspect of responsibility forms a significant part of authorship itself. Even if the legal issues related to AI authorship were resolved, then, (for

instance, by allowing AI models to *assent* to research, or allowing for proxy consent, as if the chatbot were a precocious child who was not legally able to able to sign authorship forms itself), this responsibility inherent to authorship itself is not something existing (or likely imminent) AI tools are able to achieve.

We do not strip written works of their authors once those authors perish. But a deceased author, though no longer available to explain or stand up for any aspect of their work, was once responsible for their work in a way AI cannot be. The author does not simply generate words; they write for a particular purpose—to explain to or persuade others of their ideas, for instance—and with an understanding that their professional reputation will be affected by what they say. It is in this way that authors are responsible for their work, even if they can no longer take responsibility for their work at some later date. Authors have a rationale for and understanding of what they write, and this fact about the creation of the work does not disappear once the author dies or becomes incapable of explaining their reasoning through incapacitation. AI, on the other hand, cannot explain or justify its reasoning because it never had any reasoning to explain in the first place.

Anthropomorphic language litters discussions of how AI chatbots work: the chatbot “understands” the written word; it “decides” what text to generate in response to the prompt. But just because we use the language we are familiar with to describe in lay terms what AI tools are doing does not mean these words accurately convey what is going on (Shanahan 2024). Even calling the chatbot “it,” although not an anthropomorphization, is still a misleading and oversimplistic gloss. Calling a chatbot “it” makes it seem like “it” is one discrete thing—one specific and definable *entity*—with which we are interacting whenever we use the chatbot.

However, even if we asked the same version of ChatGPT (version 3.5, say, rather than the latest update) that was used to develop the manuscript to explain or justify its process, we will not be asking the same “entity” as created the work. Although a recent update to ChatGPT enables it to “remember” what the user requested in a previous session, there is no system-wide memory that these chatbots can use to form a coherent rationale for its outputs. And it is not just that the chatbot cannot explain its reasoning in the same way a deceased author cannot. More importantly, it cannot provide a justification because it did not in any meaningful way understand the output it generated (Bender and Koller 2020; Lake and Murphy 2023; Maggiori 2023; Mitchell 2019; Saba 2023; Shanahan 2024). Nor would this work be reproducible as one may expect from a program or statistics package.² The original product was created using statistical predictions about word sequences, and the new output in response to a prompt seeking justification or explanation would be created in the same way. There is no comprehension or reasoning involved in connecting the two outputs. What

the deceased author lost in dying was nonetheless there in the creation of the work; the chatbot never had it to begin with.

There is another reason, related to responsibility but less often discussed, that supports the case against granting AI authorship for generating text in research contexts. There's a difference between generating text and writing, in that writing is an intentional act. In a very important way, AI cannot be an author because it does not intend to write, let alone publish, the text that it generates in any meaningful sense of the term (see, for instance, Shanahan 2024; Woudenberg van, Ranalli, and Bracker 2024). In writing something for publication, however, we intend to communicate something, and we intend for it to be published. *Intentional* authorship may not be an explicit part of authorship guidelines, but this omission is likely because this intention can reasonably be assumed when someone submits a manuscript to a publisher (and this assumption can be confirmed by signing a publication agreement). Most people do not write and submit manuscripts while sleepwalking, for instance, and so this is not a common occurrence that authorship guidelines need to guard against. That intentionality is important to at least our folk understanding of authorship, though, can be seen in the ethical debates that arise whenever some incomplete work by a deceased author is discovered and published. In these cases, it is not clear that *those words* are the words the author would have intended to communicate in their finished work, let alone whether they intended them to be published at all. These ethical concerns are even more pressing when a work is posthumously published that its author had wanted destroyed, as can be seen with the recent publication of Gabriel García Márquez's novella *Until August* (Belth 2024). If intentionality is a morally important aspect of authorship and publication, then this provides additional support for why AI cannot be an author.

Therefore, we argue, even if authorship guidelines were changed to remove responsibility as a separate criterion, AI authorship (the use of AI tools to generate original text) should still be prohibited in research publication. The problems with granting AI authorship run deeper than mere compliance with the current guidelines. Even if we look beyond the question of compliance with existing guidelines, there are broader reasons to deny AI authorship in the current publishing environment. The inclusion of AI authors is incompatible with ethical publication norms and will have detrimental effects on the broader scientific ecosystem. These same considerations also support prohibiting the use of AI in ways that would merit authorship were they completed by a human, such as through generating "original" text.

6. Fair credit

AI tools cannot give proper credit to ideas and should not be scientific authors. The lack of public understanding of the training data set means

that it is impossible to properly attribute credit for the words and ideas that it generates. The artist Sarah Andersen raises a similar issue with the AI image creator StabilityAI, that is able to create novel artwork done in the style of her existing work, but the AI was trained on her own artwork without her permission and without compensating her (Ofgang 2023). The same situation exists in using ChatGPT for research articles. The chatbot's underlying AI model was trained on people's work without their explicit consent and without any credit. This lack of transparency and the inability to offer credit where it is due makes the current use of ChatGPT unethical.

The crux of the argument is that these data sets contain the work of artists and are used without their consent, credit, or compensation to produce work that they themselves now must compete with. The art-producing AI services can create artwork in the style of individual artists forcing those artists to compete with corporations using their own intellectual property that was used without their consent. The people whose work was used to train ChatGPT and other similar services are for the most part unknown and both unable to give their consent and unable to be identified to be given the credit that they are due. This is due to the limited amount of knowledge provided by the makers of AI tools about which data sets their model was trained on. Full disclosure of the training sets would at least allow people whose work has been used to request removal from new training sets and/or to request compensation.

Who deserves credit for the intellectual work produced by ChatGPT? We would argue that the intellectual credit due to ChatGPT-created works should be shared by the developers of the large language models that teach the AI systems to predict the next word in a sentence and the people whose words and ideas were used to train the model. Fair credit can be given to the algorithm makers as part of a user agreement that dictates who gets to use the AI and spells out the proper way to attribute credit for it. This attribution is pretty straightforward. The attribution for people whose work was used to train the data sets is trickier. The "move fast and break things" mind-set does not lend itself to scrupulous attention to fair credit and respect for the intellectual property of things available to scrape from the internet.

To illustrate, consider an example of an AI-chatbot trained only on New York Times publications. It can create an effective facsimile of the Times style of reporting. It then scrapes the internet for publicly available information and writes it up in the Times style and offers this product up to readers as a competitor to the New York Times. It is easy to imagine the Times as regarding this competition as illegitimate and to object to the use of its intellectual property to create its own competition. It has sued OpenAI on these exact grounds (Grynbaum and Mac 2023). Our focus, though, is on the ethical aspects of fair use and not the legal issues. Regardless of the legal outcome, the current generation of LLM-based tools do not and cannot

provide fair and accurate credit for the sources it uses in generating its output. As it stands right now, the use of ChatGPT and related services to generate text for manuscripts is unethical due to the inability of these systems to properly attribute credit for the intellectual work used to train them, and this lack of credit is incompatible with the ethical obligations of fair credit and disclosure in research publications.

7. The ecology of the publishing environment

In addition to raising ethical issues about fair credit, the use of AI also has the potential for other negative effects on the publication landscape. The real advantage in using AI-generated publications would be to increase one's odds of getting a manuscript published by dramatically increasing the number of papers a given scholar has under review (for further discussion of this problem, see, for instance, Resnik and Hosseini 2023). A numbers play of increasing the total amount of papers under review would undoubtedly increase the number of papers published overall but would create a good deal more work for researchers, journal editors and reviewers. Evolutionary biologists differentiate between *r* and *K* reproductive strategies (MacArthur and Wilson 1967). Organisms that follow an *r* reproductive strategy produce large amounts of offspring with little investment in each. Dandelions follow an *r* strategy in that they release substantial amounts of low-energy seeds into the environment with little investment in any one individual potential offspring. On the other hand, organisms that pursue a *K* strategy produce few offspring but put considerable investment in each. Elephants are a good example of organisms who produce few offspring but invest heavily in their evolutionary survival. The effectiveness of *r*/*K* strategies are determined by the salient feature of an organism's selective environments.

Currently, we would argue that non-predatory academic publishing is a kind of selective environment that favors *K*-type publication strategies. Producing substantial amounts of low-investment research is unlikely to lead to a strong publication record. Peer reviewers are happy to recommend rejection for low-quality submissions repeatedly. Even if eventually published, these types of work are unlikely to enhance one's scholarly reputation much. It is essential to invest in the quality of an article before it is submitted if one hopes to get it published. Of course, the existence of predatory publishing outlets that will publish anything for a fee already complicates the ecology of academic publishing. Some scholars are pursuing a high-volume low-quality *r*-type research program.³ Our view is that permitting AI-authored research papers would help create a publishing environment that rewards poor work like that generally found in predatory journals.

The advent of AI-assisted publication opens up the possibility of employing an *r*-based publication strategy in which AI does most of the

important work in creating a paper, thus enabling someone to increase their journal submission rate by 10. If it is easy to create a new paper each week, AI-assisted researchers could submit 50 papers a year. A couple of these are bound to get published. Of course, the selection in this instance comes at the very real cost of dramatically increased editor and peer-reviewer time. This dramatic cost of reviewer and editor time is another reason to suggest that it is best not to allow AI coauthors. It is possible that the use of AI tools will not increase one's efficiency in producing research manuscripts, but if that is the case then we would expect that AI usage would decline and questions about its morality would become moot.

Adopting AI-driven peer review is no solution to this problem either. AI models and tools lack the expertise to serve as an effective peer reviewer. AI tools may be able to generate quality text most of the time, but they are not even close to judging whether a given submission makes a substantial contribution to a scientific field. The lack of domain expertise dooms this kind of approach. In addition, LLMs have well-known biases (Mollaki 2024; Resnik and Hosseini 2023). The training sets are largely drawn from the internet and thus exhibit the biases common to online discourse. AI reviewers are likely to discriminate against marginalized groups, perpetuating injustices directly. Finally, peer reviewing is a confidential process and there is no way to know what an LLM is going to do with any submitted text. For these reasons, LLMs should not be used for peer review and the burdens placed on the peer review system by the widespread adoption of AI-created publications suggest that it would be wrong to embrace their incorporation into science by allowing them to act as coauthors (whether officially credited as such or not).

8. Justice

There is also a broader publication ethics problem with allowing people to submit AI-generated or AI-assisted papers for publication because it makes an already crowded journal environment even more crowded for researchers needing publications for career advancement (Resnik and Hosseini 2023). The basic machinery of publication that allows people to start and build academic careers is increasingly strained by pressures on the publication system. This fact suggests an additional ethical concern about using AI to generate or produce research works; namely, that its use creates a competitive disadvantage for people unwilling or unable to use AI in their own work. If we assume that the primary reason to use AI is to create research works more quickly and efficiently, it makes sense that people who develop AI-assisted research programs will submit more papers than people who do not. This imbalance will disadvantage people and make it harder to maintain a non-AI-assisted academic future.

An analogy to performance-enhancing drugs in sports is apt. The existence of performance-enhancing drugs makes each competitor decide whether they want to cheat by taking banned substances to improve their performance. The existence of ChatGPT-enhanced research papers forces junior scholars to face a similar dilemma- do what it takes to compete or lose out to cheaters. This point is independent of the issue of AI-assisted publications overwhelming the precarious academic publishing system. Here, the problem is that some individuals choosing to aid their research by using AI tools puts everybody else at a competitive disadvantage, which creates an injustice in this context.

Of course, this is not to say every use of AI in research and manuscript preparation is wrong and ought to be prohibited by journals. It is certainly plausible that other uses of AI could be permissible and could even help rectify other injustices in publication. For instance, because most leading academic journals are published in English, non-native English speakers and those who do not speak English often have a competitive disadvantage when seeking publication in these journals and likely have to put in additional work that native English speakers do not when preparing their manuscripts. People whose primary language is something other than English are often told by journals to have their manuscripts checked by a native English speaker before submission, but not everyone has ready access to someone able and willing to provide this assistance. Allowing the use of an AI tool to improve the readability of an already written (human-generated) paper, or to translate an original paper into English, could be perfectly acceptable under certain conditions and could help level the playing field for researchers across the globe (Hosseini, Rasmussen, and Resnik 2023).⁴ Journals could permit the use of AI for improving manuscript readability while still prohibiting its use in the generation of original text, as we argue they should.

Some may argue that an AI-enhanced future is already inevitable, and the ethical debate should be about how to best mitigate the harms caused by the adoption of AI resources in academic research. It is difficult to restrict the use of new products that create significant advantages for those who use them in any context. Why might the academic/research publication be any different? As an aside, let us first suggest that the proliferation of AI-driven research is in no way inevitable; at least not in AI's current form. Tools like ChatGPT seem best suited for producing mediocre text, but generating mediocre text is not a great way to get any one particular paper published. Most reputable peer-review journals reject more papers than they accept by a large margin. A surefire way to get a paper rejected is to submit a paper with merely mediocre writing.

9. Deterrence

Hosseini, Rasmussen, and Resnik (2023) argue that the use of generative AI should be permitted and cited because its use is impossible to detect, and it is better to provide a pathway for it to be used and acknowledged as opposed to being driven underground. We generally agree with this approach but are hesitant to cede this much ground to the idea that the use of generative AI in scholarly publications is a good career move in the long term. Specifically, we argue that it is possible to deter researchers from using AI tools for generating substantial amounts of text for published papers. It may be pragmatically advantageous to manage AI usage by allowing it when properly referenced, but this is an opportunity to stand on principle and ban the use of generating text by AI means altogether. A consensus policy by leading journals to treat substantial text generation by AI tools as using another's words without proper attribution—i.e., plagiarism—would establish a clear ethical norm.

In addition, the jury is still out on the wisdom of using AI as a text generator for career success. Nobody knows how future hiring and tenure committees will assess the weight of AI-assisted publications. Publications are a means to build a reputation. It is currently unclear if generating a substantial number of AI-assisted publications will add to one's academic reputation if people know that AI was used to generate text. It could be seen as a career negative.

It is true that current tools do not accurately identify text generated by AI, but we do not think that this is likely to be true in the future. Plagiarism detection is a billion-dollar industry that is in serious danger of being made immediately obsolete by this new technology. For example, the plagiarism detection software company Turnitin.com recently sold for 1.75 billion dollars (Johnson 2019). A plagiarism detection company that cannot detect the use of AI to plagiarize adds little value in the current context. Many economic resources will be dedicated to the problem of AI detection in the future. One could imagine a bit of a technological arms race in which new forms of AI-generated text would be undetectable by current tools, but this need would lead to the continual development of better and better AI detection tools.

It is likely that there will never be an accurate real-time AI detection tool. However, in scholarly publishing, deterrence can be achieved by an accurate backward-looking detection mechanism. For example, recent work on the explosive growth of the word “delve” since the release of ChatGPT suggests that researchers will be able to piece together unusual language patterns that will be revealed as telltale markers of LLM-generated text (Bonyhady 2024). If a researcher publishes a paper using generative AI in a way that would normally be considered plagiarism, i.e., cutting and pasting large chunks of AI-generated text into a paper without citation or other attribution, and it is

discovered later, the consequences would be catastrophic for one's career. Research misconduct has no ethical statute of limitation in that credible discoveries of past misconduct have a way of derailing even the most established academic careers, as the recent resignation of the president of Stanford University illustrates (Tanne 2023). The fact that even very distant research misconduct has the possibility to wreck a successful research career and the high likelihood of effective backward-looking AI detection tools suggest that meaningful deterrence is possible. To do so would require consensus on which uses of AI count as research misconduct and the effective communication of this consensus to research communities.

Furthermore, and perhaps more importantly, authorship guidelines also play a role in shaping and reinforcing the research community's norms. Because of this, the construction of these guidelines should not be limited to creating criteria we can readily enforce but should also be driven by considerations of what we value about authorship and in the research community. Using AI to develop manuscripts is certainly more efficient than the painstaking effort of natural, human writing, but is efficiency all we want from research and publication? Do we want as many papers as possible (no matter how these papers are produced) or do we want good research and good researchers?

The problem with the research landscape does not seem to be that we do not have enough published papers available. Instead, one of the more pressing problems is the “publish or perish” environment that leads some researchers to take unethical or ethically questionable shortcuts in research and writing. Allowing AI-developed manuscripts would likely do nothing to ameliorate this situation and may instead make it worse, given that it could further raise expectations about what a researcher can accomplish in a limited amount of time (Resnik and Hosseini 2023). This would then lead to a greater quantity of questionable research that must be sifted through to find quality information, not less.

In addition to the potential impacts on publications, we should also think carefully about the impact of AI-generated manuscripts on researchers. Some may happily envision a world where humans work as supervisors or fact-checkers for AI-produced manuscripts (see, for instance, Zohny 2023), but is this a world we should want? Anyone who writes knows how much more difficult it is to start back up after taking some time away. What we stop doing we quickly become less able to do. And writing is not just the mechanical activity of producing words, but a way of working through our reasoning process; writing is itself thinking. Although we may gain time by relying on AI to write up our work, we should ask whether we are gaining or losing something more valuable in the process. If we demote ourselves to supervising the work being done by AI, we may end up weakening our ability to critically evaluate research (and thus to function as competent supervisors

of AI's work) along with our capacity for creative idea generation. Unless the time gained is used in a much more rewarding way that compensates for these losses (and is not just used to supervise yet more AI-generated texts), it is hard to see how this will be an appealing career path in the future. As an editorial on AI and authorship in the journal *Biological Psychology* asks, "If the use of language in meaningful texts counts among the most advanced skills of human culture, why should people not desire ownership or take pride in their products?" (Ritz 2023, 108590) If we outsource this "meaning-making" to AI, which cannot itself *mean* anything (Bender and Koller 2020), we risk impoverishing both the research and ourselves.

It is possible, of course, that the choice between the quantity and quality of papers is a false dichotomy; there may very well be ways to use AI to have both more and better research, with no other harms to the research community or the world at large. But until we are sure that this is so and have a clear conception of (and argument for) when—in what specific contexts and for which uses—the possible benefits of using AI in research outweigh the risks and harms, we should be cautious about normalizing its use. Although in this paper we have focused on potential harms to the research and researchers, the environmental harms posed by the development and use of AI tools are significant (Strubell, Ganesh, and McCallum 2019; Tamburrini 2022) and should also factor into our considerations. To employ AI wisely, we need to reflect on what we value in research and how and when AI could be used *responsibly* in furtherance of these values. We should not let the speed and seeming inevitability of AI development keep us from this careful deliberation about what we want the future of research and publication to look like.

10. Conclusion

This paper argues that using current forms of AI to generate text and ideas for academic research papers should not be rewarded with an authorship byline in the current research environment. There are considerable problems with their ability to take responsibility in any meaningful sense, which disqualifies them according to most authorship guidelines. Even if we consider the question removed from the guidelines, we argue that they still should not be used because they cannot give proper attribution, put undue stress on peer reviewers and are unjust to researchers who forgo their use. Additionally, their adoption is not inevitable. People can and should resist adopting and accepting the adoption of AI in academic publication if they so choose. Refusing to peer review AI-generated manuscripts is an easy place to start. Finally, we should question the idea of the production of research for publication's sake. Some forms of human endeavor are valuable as an

expression of our shared humanity. As such, we should not adopt AI authorship in the name of mere efficiency.

Notes

1. Although perhaps an argument could be made that this would be justified in certain instances, that is a separate question from the one of whether the chatbot itself can be held responsible for what it produces.
2. Thank you to Bharath Kandula for raising this helpful point.
3. Thank you to an anonymous peer-reviewer for raising this point.
4. We thank an anonymous reviewer for raising this point.

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